

# EESSI

EUROPEAN ENVIRONMENT FOR
SCIENTIFIC SOFTWARE INSTALLATIONS

3 Sept 2026

https://github.com/EESSI/meetings/wiki

# This meeting is being recorded

- Be aware: If you speak up, you will be in the recording
- Feel free to raise questions via Zoom chat instead
- We'll do informal Q&A at the end (after stopping recording)
- Recording will be made available via EESSI YouTube channel:

  https://www.youtube.com/@eessi_community

# Agenda



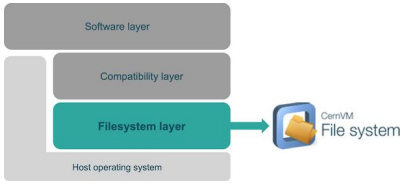

- Quick introduction by new people
- Progress update per EESSI layer
- Update on EESSI production repository `software.eessi.io` (+ RISC-V)
- Update on EESSI test suite, build-and-deploy bot, documentation, etc.
- EasyBuild v5.4.0
- Status update of EESSI on EuroHPC systems
- Progress on NVIDIA and AMD GPU support
- EESSI governance
- Upcoming/recent events
- EESSIER Transversal EuroHPC CoE
- Q&A

# Quick introduction by new people

**New people on the call: feel free to introduce yourself!**

- Who are you, where do you work, on what?
- Why are you interested in the EESSI project?
- Are you planning to actively contribute,
  and if so, to which aspect(s) of the project?

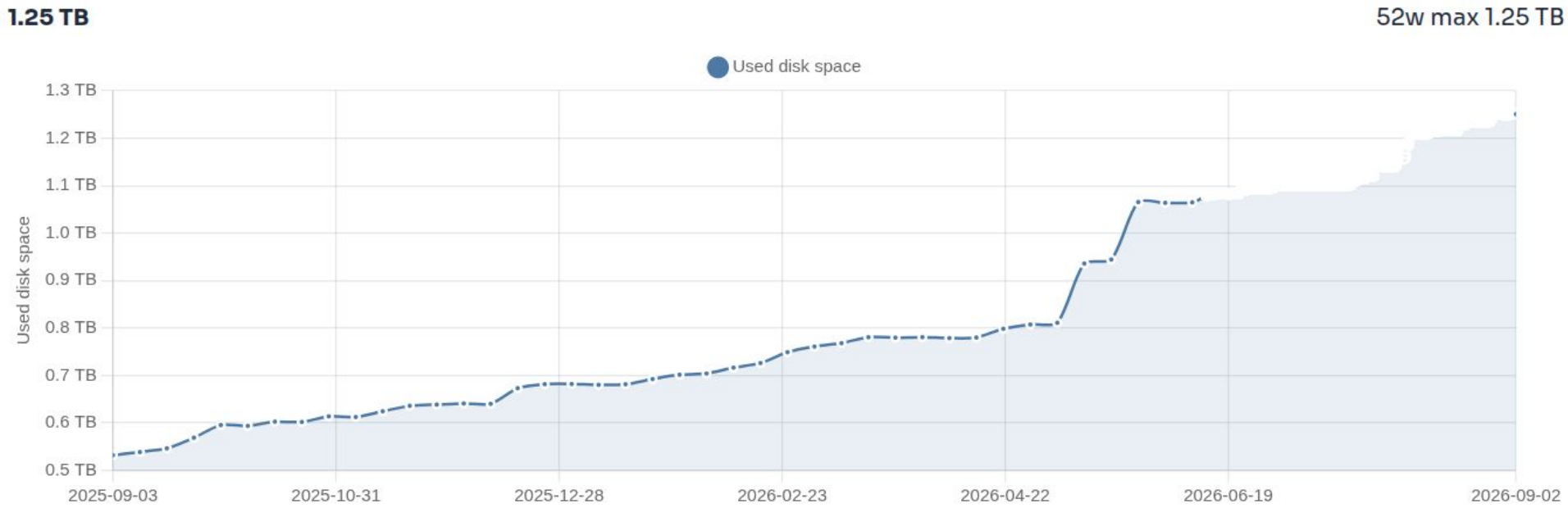
# Progress update: filesystem layer




- No issues with the CernVM-FS infrastructure, all servers currently have:
  - Latest Linux kernel version
  - CernVM-FS version 2.14.1
    **(this version includes support for partial replication!)**
- Updated the Debian 13 build container (PR #272)
- Script for publishing named snapshots (PR #276)
- Make symlink for (hidden) `EESSI/2026.06` modulefile in `/cvmfs/software.eessi.io/init/modules` (PR #277)
- Let `ingest_tarball.sh` script support custom installation prefixes (PR #278)

[Bob, Kenneth]

# Progress update: filesystem layer monitoring

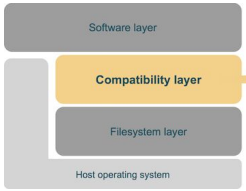



Currently ~1.25TB of disk space is required
for a full Stratum-1 mirror of all EESSI repositories

This information is now presented at https://status.eessi.io/trends.html

[Bob, Kenneth]

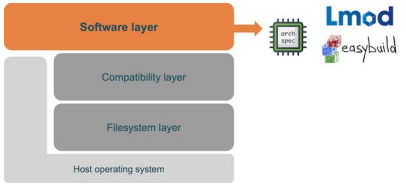
# Progress update: compatibility layer




- Compatibility layer for EESSI 2026.06 has been available for ~2 months
- Rebuilt once in the meantime
  - Remove build dependencies like CMake, Ninja, Meson
  - See https://github.com/EESSI/compatibility-layer/pull/237
  - Thanks to Bart Oldeman (The Alliance) for reporting this via issue #236

[Bob, Kenneth]

# Progress update: software layer (1/3)




Highlights of recently merged software PRs (2023.06):

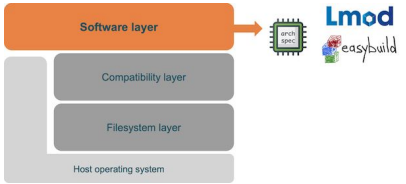
- 670 distinct software packages available; 3 merged PRs since last update meeting (2 Jul'26)

Highlights of recently merged software PRs (2025.06):

- 514 distinct software packages available; 23 merged PRs since last update meeting (2 Jul'26)
- More top level applications available:
  Siesta 5.4.2 CUDA 12.6.0 (PR #1506), huggingface-hub 0.34.4 1.2.4 (PR #1511), CPU deps for HIP and RCCL (PR #1527), CUDA-samples 12.9 (PR #1541), ollama 0.24.0 w/ CUDA 12.8.0 (PR #1542), googletest-1.17.0 (PR #1503), SMC++ 1.15.4 (PR #1548), R-bundle-CRAN 2025.11 & 2026.10 (PR #1550 & PR #1551), Arrow 22.0.0 (PR #1552), R-bundle-Bioconductor 3.22 (PR #1554), EasyBuild 5.4.0 (PR #1559), IRkernel 1.3.2 (PR #1562), ESPResSo 4.2.2 w/ CUDA 12.6.0 (PR #1565), Pint 0.24.4 and Boost.MPI 1.85.0 (PR #1566), buildenv for lfoss-2025b (PR #1588), BEDTools 2.31.1 (PR #1595), Bowtie2 2.5.4 (PR #1596), HMMER 3.4 (PR #1600), STAR 2.7.11b (PR #1601), MEME 5.5.9 (PR #1603), OpenMM 8.5.2 (PR #1604), Jellyfish 2.3.1 (PR #1606)

[Kenneth, Thomas, Richard, Pedro, Bob, Lara, Caspar, Satish, Alan, Mattias, Danilo, Helena …]

# Progress update: software layer (2/3)




Highlights of recently merged software PRs (**2026.06** 🎉):

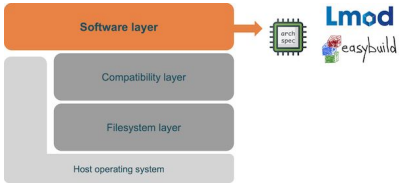
- 111 distinct software packages available
- 16 merged PRs since last update meeting (2 Jul'26)

  Mostly toolchains, etc., for now, but first top-level applications start to appear: Python-bundle-PyPI 2026.04 (PR #1578)

  New supported CPU target: Intel **Granite Rapids** 🥳: PR #1605, PR #1609, & PR #1610 (ongoing)

[Kenneth, Thomas, Richard, Pedro, Bob, Lara, Caspar, Satish, Alan, Mattias, Danilo, Helena …]

# Progress update: software layer (3/3)

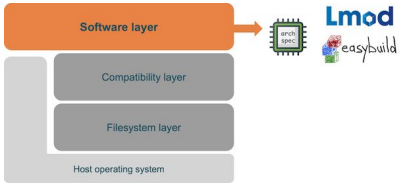



Rebuilds of existing software installations:

- EESSI 2023.06
  - Rebuild `Siesta/5.2.2-foss-2023a` (PR #1535)
  - Rebuild software that depends on ParMETIS (PR #1553)
- EESSI 2025.06
  - None!

[Kenneth, Thomas, Richard, Pedro, Bob, Lara, Caspar, Satish, Alan, Helena …]

# Progress update: software-layer-scripts



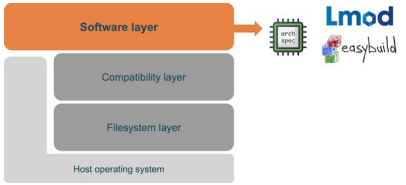

Highlights of recently merged software-layer-scripts PRs:

- Improve software-layer-script workflow (PR #147)
- Allow for compiler+accelerator specific MPI overrides (PR #231)
- Various 2026.06 related PRs (#266, #267, #279, #282, #284)
- Extend CUDA supported CC table, add CUDA and cuDNN host injections easystack for 2026.06 (PR #284)
- Clean up `dev.eessi.io`-specific code (assume it's doing a site installation) and improve tarball creation step for site installations (PR #295)

+ various other minor PRs: #235, #261, #262, #263, #269, #270, #272, #274, #275, #276, #280, #287, #288, #289, #291, #292, #296, #297

[Kenneth, Alan, Bob, Richard, Thomas, Aayush …]

# Progress update: RISC-V




RISC-V software stack being built via https://github.com/EESSI/dev.eessi.io-riscv

- See page in documentation: https://www.eessi.io/docs/repositories/dev.eessi.io-riscv/
- Available software page: https://www.eessi.io/docs/available_software_riscv/
- Installations targeting `/cvmfs/dev.eessi.io/riscv` repo
  - Same installation workflow through bot
  - Built on top of EESSI 2025.06 compatibility layer for RISC-V (available in `software.eessi.io`)
- >300 installations have already been ingested for `riscv64/generic` 🎉
  - `foss/2025a`, `foss/2025b`, and **`lfoss/2025b`** are available
- CPU target `riscv64/sifive/p550` is available
  - `foss/2025b` is already available
- Currently looking into:
  - More end-user applications (e.g. GROMACS 2026.02)
  - Going beyond `riscv64/generic` as target, also optimize for Banana Pi (RISC-V CPU w/ vector instructions)
  - **Support for `riscv64/generic` in `software.eessi.io` (for EESSI 2026.06)**

[Bob, Julian, Kenneth, …]

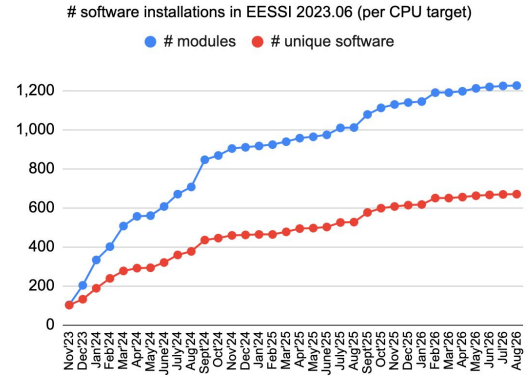
# EESSI production repository

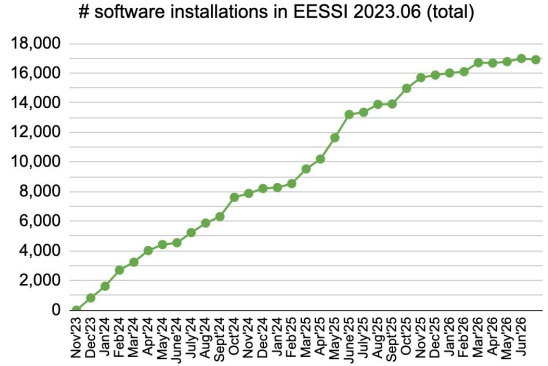
eessi.io/docs

Status update on **production-ready EESSI repository** `software.eessi.io`

Version **2023.06** is being populated with software via PRs to software-layer repo + build-and-deploy bot

- 14 fully supported CPU targets: see eessi.io/docs/software_layer/cpu_targets
- Support for NVIDIA GPUs is in place, see eessi.io/docs/gpu
- Currently: 1,226 software installations per CPU target
  (+7 in July+Aug'26)
  - 670 different software projects (+2 in July+Aug'26)
    17,115 software installations (across 14 CPU targets, +101 in July+Aug'26)
- **Currently mostly dormant, not a lot of additional software being installed in EESSI 2023.06 (old toolchains)**
- Attention is shifting to newer EESSI versions (see next slides)
- See overview of available software: https://eessi.io/docs/available_software



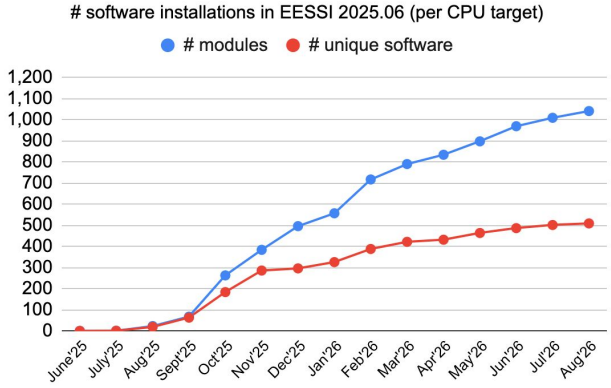
# EESSI production repository

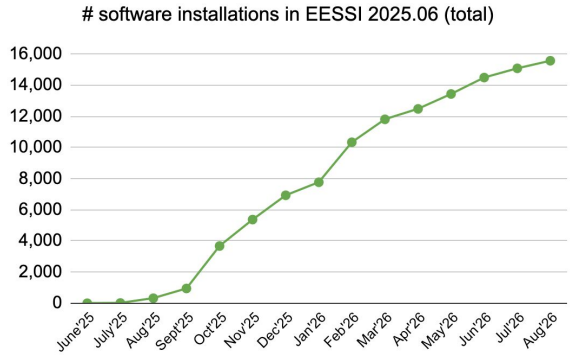
eessi.io/docs

Status update on **production-ready EESSI repository** `software.eessi.io`

Version **2025.06** is being populated with software via PRs to software-layer repo + build-and-deploy bot

- **15** fully supported CPU targets: see eessi.io/docs/software_layer/cpu_targets
  - `x86_64/amd/zen5` is additional compared to EESSI 2025.06
- Support for NVIDIA GPUs is in place, see eessi.io/docs/gpu
  - AMD GPU support is there, but not exposed yet by archdetect
- **Currently: 1,041 software installations per CPU target** (+72 in July+Aug'26)
  - **509 different software projects** (+22 in July+Aug'26)
    15,560 software installations (across 15 CPU targets, +1,078 in July+Aug'26)
- See overview of available software: https://eessi.io/docs/available_software
- Current focus:
  - Adding more software
  - Processing incoming contributions
  - GPU software installations (CUDA + ROCm)



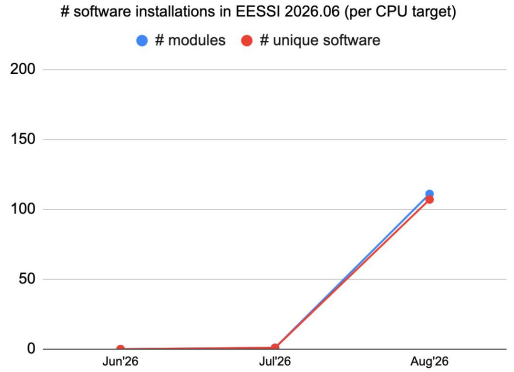
# EESSI production repository

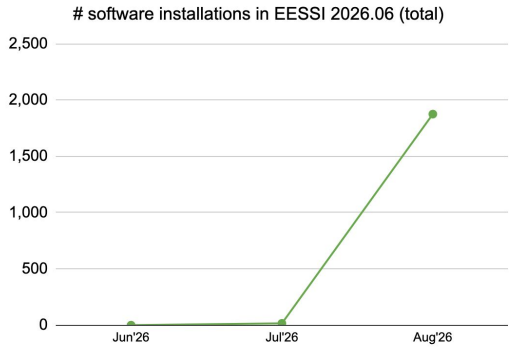
eessi.io/docs

Status update on **production-ready EESSI repository** `software.eessi.io`

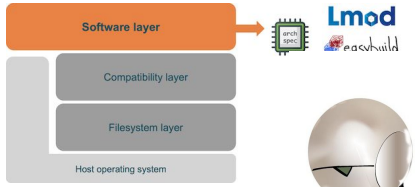
Version **2026.06** (**new!**) is being populated with software via PRs to software-layer repo + build-and-deploy bot

- **16+1** fully supported CPU targets: see eessi.io/docs/software_layer/cpu_targets
  - `aarch64/aws/graviton4` is additional compared to EESSI 2025.06
  - `x86_64/intel/graniterapids` is now also being added (WIP)
  - `riscv64/generic` is included as experimental CPU target
- **Currently: 111 software installations per CPU target** (+111 in July+Aug'26)
  - **107 different software projects** (+107 in July+Aug'26)
    1,875 software installations (across 16+1 CPU targets, +1,875 in July+Aug'26)
- See overview of available software: https://eessi.io/docs/available_software
- Current focus:
  - Adding more software
  - Catching up for `x86_64/intel/graniterapids` additional CPU target



# Bot for building + deploying software






https://github.com/EESSI/eessi-bot-software-layer + https://github.com/EESSI/filesystem-layer

Developments since bi-monthly in July

- `gitlab` [MERGED] improved test coverage (PR #379)
- `gitlab` [MERGED] Update documentation - GitLab bot setup (PR #382)
- `gitlab` [MERGED] Set minimum PyGHee version to 0.2.0 (PR #388)
- `develop` [MERGED] Report commit SHAs (PR #386)
- `gitlab` [OPEN] Add GitLab support for bot: show_config command (PR #390)
- `gitlab` [OPEN] Improve issue_comment event handling (PR #391)
- `develop` [OPEN] add convenience script to start/stop the bot (PR #392)

[Thomas, Sondre, …]

# EESSI documentation

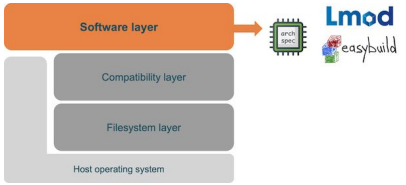
eessi.io/docs - GitHub repo github.com/EESSI/docs

Highlights:

- Disclaimer on blog posts (for performance numbers, etc.)
- Make available software page searchable by URL
- Add $EESSI_EASYBUILD_HOOKS_OVERRIDE to known variables
- Add 2026.06 to CPU targets and GPU targets pages
- EESSI Events & Training
  - Training page for hackathon  in Dublin after EuroHPC User Days
- Blog posts (https://eessi.io/docs/blog/)
  - correct RVV HPL on the RISC-V SpaceMiT X60, via EESSI
- Extensive updates for EESSI as a base for a local software stack (WIP)
- Plus other minor changes to the documentation (software overview, corrections,...)

[Lara, Pedro, Helena, Bob, Kenneth, Caspar…]

# EESSI test suite

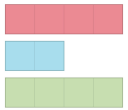



eessi.io/docs/test-suite - github.com/EESSI/test-suite

- v1.0.0 of EESSI test suite was released 29 July 2026! 🎉
  - Number of bug fixes and enhancements, see release notes
  - New and updated configurations for additional systems
  - Remove deprecated classes

[Caspar, Sam, Satish, Lara, Kenneth]

# EasyBuild v5.4.0






Released on 9 August 2026, see https://pypi.org/project/easybuild

Highlighted changes, cherry-picked for EESSI relevance:

- Significantly faster installation of easyconfigs with a lot of (build) dependencies or extensions
  - Number of times a `module load` command is run is drastically reduced…
- Improve `--bwrap` to work with r/o filesystems (overlayfs) and `--subdir-{modules,software}` (#5173)
- Fix OpenMPI and UCX easyblocks w.r.t. ROCm support (#4173)
- Add easyconfigs for update of (common) toolchains: `gmpich/2025b` (#26490), `gmpich/2026.1` (#26447), `gmpflf/2025b` (#26491), `gmpflf/2026.1` (#26492)

[Kenneth]

# Status update of EESSI on EuroHPC systems







- EESSI is currently available on 8 out 10 current online EuroHPC supercomputers:
  LUMI*, Vega, Karolina, Deucalion, MareNostrum 5, Discoverer, MeluXina, Leonardo*
  - Groundwork to make EESSI available on JUPITER is done
    (not yet in production due to problems with corrupted caches for loopback filesystem on GPFS)
  - Arrhenius is a work-in-progress (updates this month)
- EESSI is expected to be available on all EuroHPC supercomputers soon,
  thanks to EuroHPC Federation Platform (EFP)
  - Federated Software Catalogue (FSC) is officially part of the 2nd release of EFP
  - Upcoming EuroHPC AI Factories will also be required to integrate with EFP
  - See also https://docs.my-eurohpc.eu/software-catalog/system-specific/about
  - More info on EFP: https://my-eurohpc.eu

*(*) only available on-demand*

[Kenneth, Alan, Lara]

# Improved NVIDIA GPU support

Software-layer:

- Added additional CUDA packages for 2025.06
  - CUDA-samples 12.9 (PR #1541)
  - ollama 0.24.0 (PR #1542)
  - ESPResSo 4.2.2 (PR #1565)
- CUDA support added to 2026.06
  - CUDA 13.3.0 and cuDNN 9.23.0.39 (PR #1567)
  - Support for CUDA compute capability 7.0 has been dropped, replaced by 7.5
  - See https://eessi.io/docs/software_layer/gpu_targets

[Caspar, Bob]

# AMD GPU support

EasyBuild

- Higher level ROCm-based toolchains (rompi, rfoss, …) defined at easybuild-framework level
- Easyconfigs for rompi, rfoss, etc: still a work in progress
  - https://github.com/easybuilders/easybuild-easyconfigs/pull/25848
  - https://github.com/easybuilders/easybuild-easyconfigs/pull/25902

Software-layer:

- rocBLAS/4.4.0-rocm-compilers-19.0.0-ROCm-6.4.1 (WIP, PR #1549)
- Building towards HIP and RCCL (PR #1526)
- CPU-only dependencies for HIP and RCCL (PR #1527)

[Caspar, Bob]

# EESSI Steering Committee (ESC)

https://eessi.io/docs/governance

- Last ESC meeting was held on 4 Aug 2026, next one planned for 20 Oct 2026
- Most prominent agenda items discussed at last ESC meeting:
  - Status of AWS sponsored credits
    - Need to find new contact after several people have left AWS, joining HPSF may help
    - Alternatives beyond AWS (own HPC infrastructure, through QEMU, etc.)
  - Build-and-deploy bots: increased number of them, complicates management
  - Status of EESSI 2026.06
    - Additional CPU targets: yes for AWS Graviton 4 (deemed too early for Graviton 5)
    - Agreement to add riscv64/generic as additional experimental CPU target
      - Should not block deployment of software installations ready for all other CPU targets
    - Status update of software license check

[Kenneth, Bob, Thomas, Caspar, Helena]

# ISC'26 conference

- https://isc-hpc.com, 22-26 June 2026 @ Hamburg, Germany
- Half-day tutorial on EESSI, eessi.io/isc26-tutorial
- Birds-of-a-Feather sessions on both EESSI & EuroHPC Federation Platform (EFP)
- Detailed write-up available on EESSI blog: https://eessi.io/docs/blog/2026/06/26/eessi-at-isc26












[Kenneth]

# EESSI hackathon @ EuroHPC User Days

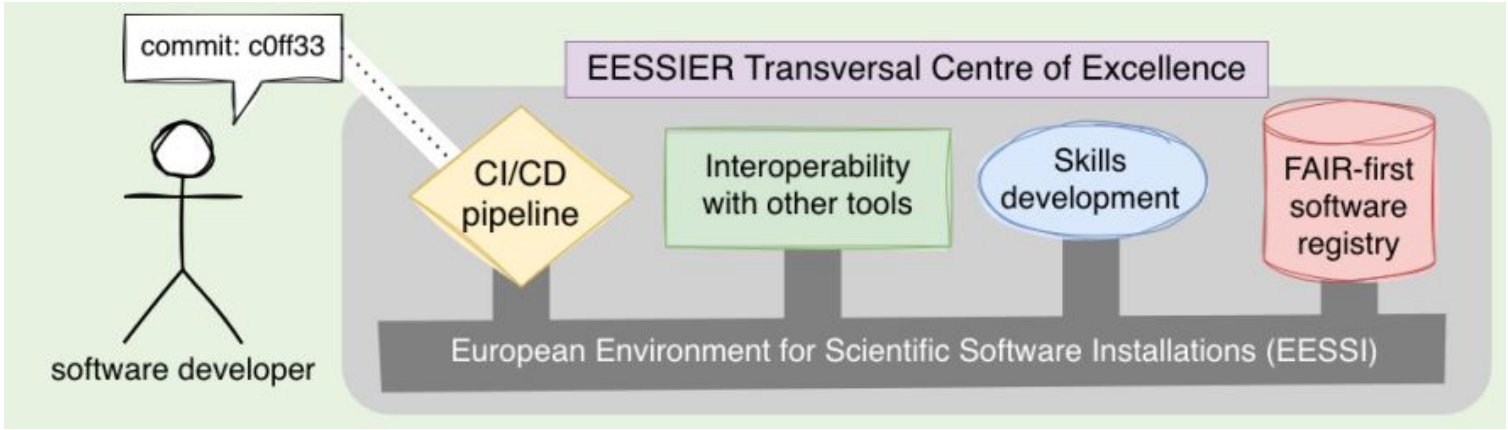
- New EESSI hackathon being organised in Dublin on Fri 25 Sept 2026
- In wake of EuroHPC User Days (22-24 Sept), https://userdays2026.eurohpc-ju.europa.eu
- Repeat of successful hackathon we did in Copenhagen in 2025, see https://eessi.io/docs/blog/2025/10/20/eurohpc-user-day-2025
- Currently 17+5 people registered to attend in-person (+ 12 online)
- Focus will be adding software to EESSI, with lots of hands-on work

https://eessi.io/docs/training-events/2026/hackathon-eurohpc-user-days

[Kenneth, Alan, Lara, Thomas, Helena]

# EESSIER Transversal EuroHPC CoE

- EESSI is the focus of a new Transversal EuroHPC Centre-of-Excellence
- EESSIER project planned to start in early 2027
- Follow-up of (technical pillar of) MultiXscale EuroHPC CoE (which ends Dec'26)
- More details to come…for now we just share the good news



# EESSI mailing list eessi@list.rug.nl

- Low-frequency mailing list
  - Announcing bi-monthly meetings
  - Important updates and announcements
- To (un)subscribe, send an email to listserv@list.rug.nl with the following body:
  - Subscribe:

    ```
    SUBSCRIBE EESSI Firstname Lastname
    ```

  - Unsubscribe:

    ```
    UNSUBSCRIBE EESSI
    ```

# Next EESSI update meetings

- EESSI update meetings are **bi-monthly** (odd months)
- First Thursday of the month at 14:00 CE(S)T, only in odd months
  - iCalendar URL for calendar integration
- Next meetings:
  - Thu 5 Nov 2026 14:00 CET (13:00 UTC)
  - Thu 7 Jan 2027 14:00 CET (13:00 UTC)
  - Thu 4 Mar 2027 14:00 CET (13:00 UTC)